



Institutional trading and satellite data

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ABSTRACT

This study investigates the impact of satellite data availability on the market timing abilities of institutional investors. By analyzing the introduction of satellite coverage for major retailers, this study finds that access to satellite data significantly enhances the return predictability of daily institutional trading. This improvement is more pronounced for stocks with severe information asymmetry and is driven by non-hedge funds. These results suggest that institutions, especially non-hedge funds, actively adjust their trading positions on a daily basis in response to satellite data.

1. Introduction

The rapid advancement of computer science is revolutionizing the information-processing methods of the finance industry. Sophisticated investors now leverage substantial amounts of alternative data, such as satellite imagery, to enhance their investment decisions. Katona et al. (2024) reveal that short sellers became more informed regarding forthcoming quarterly reports of retailers with satellite coverage following the introduction of satellite data. Additionally, Bonelli and Foucault (2024) ascertain that the release of satellite data negatively affected the stock-picking abilities of mutual fund managers, who were deficient in the requisite skills to process the data effectively. Prior research has focused on quarterly changes in the holding positions of sophisticated investors to assess the impact of satellite data on their stock-picking abilities. However, it is crucial to recognize that sophisticated investors can adjust their positions at any time during a quarter to more accurately align with their predictions for the forthcoming reports, reflecting satellite data in real time. Satellite data's impact on the market timing performance of sophisticated investors has largely been overlooked in the literature. Therefore, this study aims to address this gap by examining the return predictive power of daily institutional trading for stocks covered by satellite data in the periods before and after its introduction.

This study examines satellite data's impact on institutional trading strategies by first identifying the exact date when RS Metrics commenced the provision of state-level parking lot traffic data extracted from satellite imagery in the U.S. market. RS Metrics is recognized as the first vendor to introduce nearly real-time daily data feeds in 2009, allowing sophisticated investors, such as institutional traders, to incorporate this daily data into their trading strategies (see Katona et al. (2024) for more details). This study restricts the sample period to one month before and after the introduction of satellite data for major retailers covered by the data to focus on the immediate response of institutions to this data's availability. Furthermore, this study employs daily institutional trading records obtained from ANcerno Ltd., a company that gathers transaction records from institutional clients who utilize its trade cost analysis services (see Puckett and Yan (2011) for more details). Despite ANcerno's dataset including diverse equity transactions from a large sample of institutions, its sample period is limited to the first quarter of 2012. Owing to these restrictions from the RS Metrics and ANcerno datasets, the sample period starts from the first quarter of 2009 to the first quarter of 2012. Notably, this study's final sample includes 779 stock-day observations for 19 major U.S. retailers.

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Based on the final sample, this study finds that institutional trading more accurately predicted daily returns for stocks that were covered by satellite data following its release, with no evidence of reversal observed. This result suggests that institutions actively incorporated satellite data into their day-to-day trades, at least within the first three months of its availability. Additionally, the impact of satellite data was more pronounced for stocks characterized by severe information asymmetry, including those with low market capitalization, low analyst coverage, and low institutional ownership. Interestingly, this study reveals that non-hedge fund trades, rather than hedge fund trades, primarily drive this effect.¹ Finally, a difference-in-differences analysis is performed, comparing the covered stocks to matched stocks without satellite coverage during the sample period, and the results are qualitatively consistent with the main findings.

This study makes two significant contributions to existing literature. First, it enhances the rapidly expanding body of research on satellite data's impact on financial markets (Zhu, 2019; Mukherjee et al., 2021; Bonelli and Foucault, 2024; Katona et al., 2024). To the best of my knowledge, this is the first study to provide evidence of satellite data's impact on the market timing activities of sophisticated investors. Particularly, it demonstrates a significant improvement in the return predictability of institutional trading for stocks covered by satellite data after the introduction of such coverage. Second, by analyzing the period surrounding the introduction of satellite data, this study reveals that institutions respond almost immediately to the release of new alternative data. Huang et al. (2020) show that institutional investors promptly respond to news releases concerning corporate topics. This study determines that institutional investors also respond to the release of satellite data, thus augmenting the existing literature.

2. Literature review and research hypothesis

Numerous studies have shown that different types of alternative data, including social media activity, geolocation data, employee satisfaction reports, and satellite imagery, provide significant insights into stock returns and firms' fundamentals. Froot et al. (2017) reveal that managers distort their disclosures when insider trading opportunities arise based on proprietary consumer activity data. Similarly, Zhu (2019) shows that big data's introduction reduced the information asymmetry between corporate insiders and sophisticated investors, improving market efficiency. However, Dessaint et al. (2024) contend that alternative data is more effective for predicting short-term performance rather than long-term prospects, though Chang and Da (2022) provide evidence supporting big data's long-term predictive power. At a macro level, Mukherjee et al. (2021) demonstrate that crude oil prices exhibit heightened sensitivity to government announcements when satellite estimates of U.S. oil inventories are less reliable due to weather conditions.

Several studies have examined the role of alternative data in financial markets by considering the trading behavior of sophisticated investors. Katona et al. (2024) investigate the use of satellite data, such as store-level car counts, to inform profitable trading strategies ahead of earnings announcements. Their results indicate that satellite data increases informed trading activity, including short selling, while reducing liquidity. Bonelli and Foucault (2024) extend this line of research by demonstrating that the availability of satellite data diminishes the stock-picking abilities of mutual fund managers, underscoring the challenges traditional investors experience in processing alternative data. These studies have primarily focused on quarterly changes in the holdings of sophisticated investors to evaluate satellite data's impact on their stock-picking abilities. However, it is critical to recognize that these investors can adjust their positions at any point during a quarter to better align with their forecasts for forthcoming reports. Thus, this study contributes to the literature by investigating the impact of satellite data availability on the market timing ability of daily institutional trading.

According to Katona et al. (2024), satellite data offers valuable insights for anticipating quarterly retailer performance. Prior to the availability of satellite imagery, this information was primarily accessible to local investors, whereas non-local investors could only obtain it at a substantial expense (Kang et al., 2021). The introduction of satellite data enables market participants, irrespective of their location, to acquire this information at a lower cost and with enhanced accuracy. The reduction in information acquisition costs allows sophisticated investors — such as institutional investors capable of processing satellite data — to gain more insights into retailers' stock returns (Grossman and Stiglitz, 1980). Consequently, institutions having access to this data are likely to possess more accurate information about future stock prices. Therefore, this study proposes the following hypothesis:

Hypothesis: Satellite data availability enhances the predictive power of daily institutional trading for the stock returns of major retailers covered by the satellite data.

3. Data and summary statistics

This study compiles its sample by integrating three distinct databases: CRSP, ANcerno, and RS Metrics. Specifically, ANcerno data is linked to CRSP using the eight-digit CUSIP codes every day, and RS Metrics is merged with CRSP using stock symbols. The study employs (1) daily CRSP data for stock returns, Amihud (2002) illiquidity measure (Amihud), relative bid-ask spreads (Spread) and market capitalization (MarketCap); (2) ANcerno data for stock-day institutional order imbalances; and (3) RS Metrics data for the starting date of satellite coverage for 19 major U.S. retailers. Furthermore, it measures institutional ownership (InstOwn) from Thomson Reuters's 13F data and analyst coverage (Analyst) from I/B/E/S. It uses risk-adjusted mid-quote returns (Mid-Q Returns) with respect to the Fama and French (1993) factors and Carhart (1997) momentum factor, thereby emphasizing the idiosyncratic

¹ This study aims to exercise caution in interpreting this finding, considering that ANcerno's trade records do not fully capture all institutional trades (Puckett and Yan, 2011). The result does not imply that hedge funds do not utilize satellite data in their daily trading; instead, it suggests that non-hedge funds may be more active in employing such data.

Table 1

Summary statistics.

This table shows the summary statistics for the period from one month before to one month after the starting date of satellite coverage for each covered stock. The sample includes 19 major retailer stocks with data available from RS Metrics, ANcerno, and CRSP. AN represents the order imbalance in ANcerno data. AN HF and AN NHF refer to hedge fund and non-hedge fund order imbalances, respectively, in ANcerno data. Amihud is the Amihud (2002) illiquidity measure. Spread is the daily relative spread. InstOwn refers to institutional ownership from Thomson Reuters's 13F data. Analyst represents analyst coverage from I/B/E/S data. MarketCap refers to market capitalization. Mid-Q Return is the daily risk-adjusted mid-quote stock return with respect to the Fama and French (1993) factors and the Carhart (1997) momentum factor.

	# Days	# Stocks	Average	Std Dev	Minimum	Median	Maximum
AN	41	19	−0.004	0.066	−0.739	0.000	0.492
AN HF	41	19	0.000	0.001	−0.021	0.000	0.004
AN NHF	41	19	−0.004	0.066	−0.739	0.000	0.492
Amihud	41	19	0.027	0.044	0.001	0.008	0.208
Spread	41	19	0.043	0.034	0.004	0.031	0.329
InstOwn	41	19	0.727	0.140	0.296	0.766	0.960
Analyst	41	19	18.837	5.520	7.000	19.000	28.000
MarketCap	41	19	22.949	44.356	0.928	8.825	211.856
Mid-Q Return	41	19	0.000	0.015	−0.057	0.000	0.098

aspect of price movement.² The study examines the immediate response of institutions to the availability of satellite data by analyzing one month before and one month after the starting date of satellite coverage for each covered stock. The final sample comprises 779 stock-day observations from 2009–2012.

Table 1 provides the summary statistics. First, three versions of the ANcerno institutional order imbalance are presented. AN denotes the order imbalance in ANcerno trades, defined as the difference between buyer execution shares and seller execution shares in ANcerno data for a stock on the same day, scaled by the number of shares outstanding. Subsequently, statistics for the ANcerno imbalance from hedge funds (AN HF) and non-hedge funds (AN NHF) are reported. This study identifies hedge funds within the ANcerno data, following Agarwal et al. (2013), and then separately aggregates hedge fund and non-hedge fund order imbalances in ANcerno data.³ These three types of the ANcerno order imbalance have close-to-zero means with negative skewness. Lastly, descriptive statistics for control variables are presented. The average values of Amihud, Spread, InstOwn, and Analyst are 0.027, 0.043, 0.727, and 18.837, respectively. MarketCap is 22.949 million dollars on average, and the mean and median values for Mid-Q Returns are both recorded as 0.000.

4. Results

Satellite data offers meaningful insights into upcoming quarterly reports (Kang et al., 2021; Bonelli and Foucault, 2024; Katona et al., 2024). With access to this data, institutions possess more accurate information regarding future stock prices. Therefore, this study hypothesizes that daily institutional trading exhibits a stronger predictive power for the stock returns of major retailers covered by the satellite data after its release. To test this hypothesis, this study utilizes a panel regression model as follows:

$$\text{Mid-Q Return}_{i,d} = \alpha_i + \alpha_w + \sum_{k=1}^5 \beta_k^1 \text{AN}_{i,d-k} \times \text{After}_{i,d} + \sum_{k=1}^5 \beta_k^2 \text{AN}_{i,d-k} + \beta^3 \text{After}_{i,d} + \theta \mathbf{X}_{i,d} + \epsilon_{i,d}, \quad (1)$$

where i denotes a stock covered by satellite data, d denotes a trading day within the estimation period from one month before and one month after the start of satellite coverage, and “After” equals 1 after the initiation of satellite coverage and 0 otherwise. $\mathbf{X}$ is a vector of lagged control variables which includes InstOwn, Analyst, MarketCap, Amihud, Spread, and Mid-Q Returns. Additionally, firm fixed effects (α_i) and week fixed effects (α_w) are added to control for firm and intertemporal variations, respectively.

The results presented in Table 2 indicate a significant and positive relationship between the introduction of satellite coverage and the return predictive power of institutional trading. The estimated coefficients on the lagged interaction terms, β^1 , are all positive, with the second, third, and fifth lagged terms exhibiting statistical significance. Contrastingly, the estimated coefficients on β^2 are negative and significant at the second, third, and fifth lagged terms. The results indicate that satellite data availability significantly enhanced the return predictability of daily institutional trading for the major retailer stocks, whereas its predictive power was limited before the introduction of satellite data, possibly due to an information disadvantage compared to local investors (Kang et al., 2021).

Table 2 Column (2) presents the estimated coefficients for the five-day aggregated institutional trading (Sum(AN)) and its interaction term (Sum(AN)×After). The results presented in Table 2 indicate that the five-day aggregated institutional trading is consistent with the results for the five daily institutional trading lags. Owing to space constraints, this study reports only the five-day aggregated institutional trading results.

² In untabulated robustness tests, this study confirms that all the results remain consistent regardless of the selected type of stock returns including stock price returns or risk-adjusted stock price returns.

³ This study's sample comprises 76 hedge funds and 403 non-hedge funds, all of which traded in the 19 major retailer stocks during the sample period. Hedge funds and non-hedge funds are distinguished using ANcerno's “clientcode”, a key variable that uniquely identifies trades from different institutions (Hu et al., 2018; Huang et al., 2020).

Table 2**Return predictability.**

This table presents ordinary least squares regression results for Eq. (1). All variables are the same as defined in Table 1. The sample includes 19 major retailer stocks with data available from RS Metrics, ANcerno, and CRSP. All coefficient estimates are multiplied by 100. Corresponding *t*-statistics are reported in parentheses. ***, **, and * indicate statistical significance at 1, 5, and 10 percent level, respectively.

	(1)	(2)
$AN_{d-1} \times After_d$	1.032 (0.55)	
$AN_{d-2} \times After_d$	3.546* (1.88)	
$AN_{d-3} \times After_d$	5.739*** (3.04)	
$AN_{d-4} \times After_d$	1.263 (0.66)	
$AN_{d-5} \times After_d$	3.855** (2.01)	
AN_{d-1}	0.414 (0.33)	
AN_{d-2}	-5.070*** (-4.38)	
AN_{d-3}	-2.503** (-2.14)	
AN_{d-4}	-0.983 (-0.85)	
AN_{d-5}	-2.503** (-2.24)	
$Sum(AN)_{d-5,d-1} \times After_d$		0.030*** (4.19)
$Sum(AN)_{d-5,d-1}$		-0.021*** (-4.42)
$After_d$	-0.005 (-0.88)	-0.005 (-0.89)
$InstOwn_{d-1}$	-0.022 (-0.38)	-0.035 (-0.64)
$Analyst_{d-1}$	-0.001 (-0.62)	-0.001 (-0.49)
$MarketCap_{d-1}$	-0.004*** (-4.08)	-0.004*** (-3.95)
$Amihud_{d-1}$	0.386*** (2.85)	0.384*** (2.82)
$Spread_{d-1}$	-0.033* (-1.83)	-0.031* (-1.72)
Mid-Q Return $_{d-1}$	-0.164*** (-4.36)	-0.162*** (-4.33)
Mid-Q Return $_{d-2}$	-0.008 (-0.22)	-0.028 (-0.77)
Mid-Q Return $_{d-3}$	-0.074** (-2.01)	-0.061* (-1.68)
Mid-Q Return $_{d-4}$	-0.074* (-1.92)	-0.071* (-1.85)
Mid-Q Return $_{d-5}$	0.005 (0.14)	0.002 (0.04)
Firm Fixed Effect	Yes	Yes
Week Fixed Effect	Yes	Yes
R-Square	0.285	0.268
Num of Observations	779	779

Table 3
Return predictability in a longer horizon.

This table reproduces Table 2 to examine the return predictive ability of institutional trading across different estimation periods. For example, the column labeled Three Months (d-63, d+63) reports the coefficients estimated for the period spanning three months before to three months after the starting date of satellite coverage for each covered stock. All variables are the same as defined in Table 1. The sample includes 19 major retailer stocks with data available from RS Metrics, ANcerno, and CRSP. All coefficient estimates are multiplied by 100. Corresponding *t*-statistics are reported in parentheses. ***, **, and * indicate statistical significance at 1, 5, and 10 percent level, respectively.

	One month (d-21, d+21)	Two months (d-42, d+42)	Three months (d-63, d+63)
Sum(AN) _{d-5,d-1} ×After _d	0.030*** (4.19)	0.011** (2.04)	0.009** (2.07)
Sum(AN) _{d-5,d-1}	-0.021*** (-4.42)	-0.008** (-2.41)	-0.004* (-1.67)
After _d	-0.005 (-0.89)	-0.003 (-0.96)	-0.003 (-1.42)
InstOwn _{d-1}	-0.035 (-0.64)	0.004 (0.12)	-0.009 (-0.40)
Analyst _{d-1}	-0.001 (-0.49)	0.001 (0.59)	0.001 (1.20)
MarketCap _{d-1}	-0.004*** (-3.95)	-0.002*** (-3.09)	-0.002*** (-4.26)
Amihud _{d-1}	0.384*** (2.82)	0.247*** (3.52)	0.103** (2.10)
Spread _{d-1}	-0.031* (-1.72)	0.001 (0.08)	-0.013 (-1.30)
Mid-Q Return _{d-1}	-0.162*** (-4.33)	-0.071*** (-2.67)	-0.066*** (-3.07)
Mid-Q Return _{d-2}	-0.028 (-0.77)	-0.032 (-1.25)	-0.021 (-0.98)
Mid-Q Return _{d-3}	-0.061* (-1.68)	-0.040 (-1.59)	-0.016 (-0.73)
Mid-Q Return _{d-4}	-0.071* (-1.85)	-0.048* (-1.87)	-0.043** (-2.01)
Mid-Q Return _{d-5}	0.002 (0.04)	0.035 (1.32)	0.018 (0.81)
Firm Fixed Effect	Yes	Yes	Yes
Week Fixed Effect	Yes	Yes	Yes
R-Square	0.268	0.155	0.116
Num of Observations	779	1,567	2,305

Table 3 illustrates the effect of the introduction of satellite data on the return predictability of institutional trading in a longer horizon. Accordingly, Table 3 reproduces the regression model analyzed in Table 2 using different estimation periods. For instance, the column labeled “Three Months (d-63, d+63)” presents the estimated coefficients for the period spanning three months before and after the initiation of satellite coverage for each stock. Table 3 demonstrates that the results presented in Table 2 are robust when using two- or three-month estimation periods; however, the economic impact and statistical significance are weaker at longer horizons.

Table 4 presents the comparison of the return predictability on regular days with earnings announcement days. This study first identifies earnings announcement dates via the Compustat database and categorizes “Earnings” as equal to 1 for a day with quarterly earnings announcements and 0 otherwise. Table 4 replicates the regression model of 2, adding the interaction terms of Earnings and Sum(AN)×After, as well as Earnings and Sum(AN). Table 4 presents positive estimated coefficients for Sum(AN)×After, indicating that institutional trading has predictive power for major retailers’ stock returns on regular days after the release of satellite data. In Table 4, the interaction term between Earnings and Sum(AN)×After is positive and significant in the column for One Month but insignificant for Two Months and Three Months. The results show that, with the exception of the first month following the release of satellite data, there was no significant difference in the return predictability of institutional trading between regular days and earnings announcement days, likely due to institutions having already incorporated the earnings information into their positions prior to the announcements.

The introduction of satellite data provides institutional investors a greater advantage in forecasting stock performance, especially for stocks characterized by severe information asymmetry. This advantage stems from the assumption that institutional investors having access to such data are inclined to make accurate forecasts when information acquisition is deemed valuable. To test the

Table 4
Earnings announcements.

This table reproduces Table 2 to examine the return predictive ability of institutional trading on earnings announcement dates across different estimation periods. For example, the column labeled Three Months (d-63, d+63) reports the coefficients estimated for the period spanning three months before to three months after the starting date of satellite coverage for each covered stock. Earnings equals 1 for a day with quarterly earnings announcements and 0 otherwise. All variables are the same as defined in Table 1. The sample includes 19 major retailer stocks with data available from RS Metrics, ANcerno, and CRSP. All coefficient estimates are multiplied by 100. Corresponding *t*-statistics are reported in parentheses. ***, **, and * indicate statistical significance at 1, 5, and 10 percent level, respectively.

	One month (d-21, d+21)	Two months (d-42, d+42)	Three months (d-63, d+63)
Sum(AN) _{d-5,d-1} ×After _d ×Earnings _d	0.037** (2.01)	-0.030 (-0.86)	0.014 (0.61)
Sum(AN) _{d-5,d-1} ×After _d	0.025*** (3.53)	0.014*** (2.74)	0.009** (2.29)
Sum(AN) _{d-5,d-1} ×Earnings _d	-	-0.047 (-1.56)	-0.089*** (-5.85)
Sum(AN) _{d-5,d-1}	-0.021*** (-4.38)	-0.008** (-2.49)	-0.003 (-1.36)
After _d ×Earnings _d	-0.022*** (-3.58)	-0.100*** (-8.73)	-0.052*** (-5.94)
After _d	-0.004 (-0.72)	-0.004 (-1.20)	-0.003 (-1.53)
Earnings _d	-	0.095*** (10.03)	0.048*** (8.07)
InstOwn _{d-1}	-0.042 (-0.77)	0.008 (0.28)	-0.005 (-0.22)
Analyst _{d-1}	-0.001 (-0.40)	-0.000 (-0.02)	0.001 (1.01)
MarketCap _{d-1}	-0.004*** (-3.98)	-0.002*** (-3.21)	-0.002*** (-4.28)
Amihud _{d-1}	0.351*** (2.61)	0.224*** (3.32)	0.094* (1.95)
Spread _{d-1}	-0.036* (-1.94)	-0.005 (-0.42)	-0.013 (-1.36)
Mid-Q Return _{d-1}	-0.159*** (-4.29)	-0.067*** (-2.62)	-0.070*** (-3.33)
Mid-Q Return _{d-2}	-0.026 (-0.72)	-0.021 (-0.85)	-0.016 (-0.76)
Mid-Q Return _{d-3}	-0.052 (-1.43)	-0.046* (-1.87)	-0.021 (-1.02)
Mid-Q Return _{d-4}	-0.068* (-1.79)	-0.030 (-1.21)	-0.033 (-1.55)
Mid-Q Return _{d-5}	-0.001 (-0.03)	0.032 (1.28)	0.016 (0.75)
Firm Fixed Effect	Yes	Yes	Yes
Week Fixed Effect	Yes	Yes	Yes
R-Square	0.286	0.224	0.156
Num of Observations	779	1,567	2,305

hypothesis, this study utilizes institutional ownership (InstOwn), sell-side analyst coverage (Analyst), and market capitalization (MarketCap) as proxies for information asymmetry. A dummy variable, “D(Info Asym)”, is constructed for each proxy, which equals 1 if the value of the information asymmetry proxy is below the median and 0 otherwise.

Table 5 replicates the analysis presented in Table 2, incorporating the interaction terms of D(Info Asym) and Sum(AN)×After, along with D(Info Asym) and Sum(AN). Table 5 illustrates that the interaction term of D(Info Asym) and Sum(AN)×After is positive across all columns, attaining statistical significance in the columns pertaining to Analyst and MarketCap. The results suggest that institutions benefited more from the new satellite data’s additional information when dealing with less transparent stocks. Notably, the interaction terms of D(Info Asym) and Sum(AN) are negative across all columns. This implies that, in accordance with Tables 2 and 3, institutions experienced information asymmetry, possibly in comparison to local investors, in less transparent stocks before the introduction of satellite data (Kang et al., 2021).

Table 5
Subsample tests.

This table reproduces Table 2 to examine the heterogeneous effect of satellite data introduction on the return predictive ability of institutional trading, depending on the level of information asymmetry. A dummy variable is defined using the pooled median of a proxy for information asymmetry: D(Info Asym) equals one for stocks with a below-median value (high information asymmetry) and zero for stocks with an above-median value (low information asymmetry). The information asymmetry proxies are institutional ownership (InstOwn), analyst coverage (Analyst), and market capitalization (MarketCap), as indicated in each column. All variables are the same as defined in Table 1. The sample includes 19 major retailer stocks with data available from RS Metrics, ANcerno, and CRSP. All coefficient estimates are multiplied by 100. Corresponding *t*-statistics are reported in parentheses. ***, **, and * indicate statistical significance at 1, 5, and 10 percent level, respectively.

	InstOwn	Analyst	MarketCap
Sum(AN) _{<i>d-5,d-1</i>} × After _{<i>d</i>} × D(Info Asym) _{<i>i</i>}	0.017 (1.20)	0.041** (2.20)	0.041** (2.28)
Sum(AN) _{<i>d-5,d-1</i>} × D(Info Asym) _{<i>i</i>}	−0.012 (−1.21)	−0.028* (−1.88)	−0.028* (−1.88)
Sum(AN) _{<i>d-5,d-1</i>} × After _{<i>d</i>}	0.022* (1.74)	−0.003 (−0.14)	−0.003 (−0.17)
Sum(AN) _{<i>d-5,d-1</i>}	−0.014* (−1.78)	0.001 (0.10)	0.001 (0.07)
After _{<i>d</i>} × D(Info Asym) _{<i>i</i>}	−0.002 (−0.64)	0.002 (0.48)	0.001 (0.36)
After _{<i>d</i>}	−0.005 (−0.89)	−0.007 (−1.18)	−0.007 (−1.17)
D(InstOwn) _{<i>i</i>}	−0.011 (−0.54)	−0.015 (−0.72)	−0.017 (−0.81)
D(Analyst) _{<i>i</i>}	−0.019 (−1.60)	−0.023* (−1.82)	−0.022* (−1.79)
D(MarketCap) _{<i>i</i>}	−0.025 (−1.56)	−0.031** (−2.02)	−0.032** (−1.96)
Amihud _{<i>d-1</i>}	0.376*** (2.70)	0.457*** (3.43)	0.465*** (3.40)
Spread _{<i>d-1</i>}	−0.032* (−1.74)	−0.029 (−1.58)	−0.029 (−1.54)
Mid-Q Return _{<i>d-1</i>}	−0.190*** (−5.08)	−0.194*** (−5.17)	−0.193*** (−5.15)
Mid-Q Return _{<i>d-2</i>}	−0.058 (−1.57)	−0.062* (−1.69)	−0.060 (−1.63)
Mid-Q Return _{<i>d-3</i>}	−0.088** (−2.32)	−0.086** (−2.32)	−0.084** (−2.26)
Mid-Q Return _{<i>d-4</i>}	−0.099** (−2.47)	−0.094** (−2.39)	−0.092** (−2.36)
Mid-Q Return _{<i>d-5</i>}	−0.024 (−0.62)	−0.019 (−0.50)	−0.018 (−0.48)
Firm Fixed Effect	Yes	Yes	Yes
Week Fixed Effect	Yes	Yes	Yes
R-Square	0.250	0.253	0.253
Num of Observations	779	779	779

Next, this study examines the potential differences in trading behavior between hedge funds and other institutional investors. Due to substantial costs associated with the acquisition and processing of satellite data, prior research has suggested that only select hedge funds are typical clients of satellite data vendors (Katona et al., 2024). Nonetheless, there have been limited direct comparisons between hedge fund and non-hedge fund order flows around the introduction of satellite data. To address this gap, this study analyzes the return predictability of hedge fund and non-hedge fund order flows at the onset of satellite data providing local information on major retailers within the U.S. market.

Table 6 replicates the regression analysis in Table 2, separating the five-day aggregated institutional trading into the five-day aggregated hedge fund order flow (Sum(AN HF)) and non-hedge fund order flow (Sum(AN NHF)). Surprisingly, the estimated coefficient of the interaction term for hedge fund order flow is insignificant, whereas, for non-hedge fund order flow, it is positive and significant. This result suggests that non-hedge funds, rather than hedge funds, primarily influenced the improvement in return predictability of institutional trading. This result does not align with Katona et al. (2024). However, the ANcerno data does not comprehensively capture all the institutional trade records (Puckett and Yan, 2011; Jame, 2018). Therefore, the result does not

Table 6**Hedge funds and non-hedge funds.**

This table reproduces Table 2 to examine the return predictive ability of hedge fund and non-hedge fund order flow around the initiation of satellite coverage. All variables are the same as defined in Table 1. The sample includes 19 major retailer stocks with data available from RS Metrics, ANcerno, and CRSP. All coefficient estimates are multiplied by 100. Corresponding *t*-statistics are reported in parentheses. ***, **, and * indicate statistical significance at 1, 5, and 10 percent level, respectively.

	(1)	(2)	(3)
Sum(AN HF) _{<i>d-5,d-1</i>} ×After	0.290 (0.65)		−0.063 (−0.14)
Sum(AN HF) _{<i>d-5,d-1</i>}	−0.067 (−0.26)		0.147 (0.57)
Sum(AN NHF) _{<i>d-5,d-1</i>} ×After		0.030*** (4.20)	0.030*** (4.07)
Sum(AN NHF) _{<i>d-5,d-1</i>}		−0.021*** (−4.43)	−0.022*** (−4.46)
After	−0.005 (−0.77)	−0.005 (−0.89)	−0.005 (−0.87)
InstOwn _{<i>d-1</i>}	−0.050 (−0.93)	−0.035 (−0.63)	−0.036 (−0.65)
Analyst _{<i>d-1</i>}	−0.001 (−0.38)	−0.001 (−0.49)	−0.001 (−0.47)
MarketCap _{<i>d-1</i>}	−0.004*** (−4.22)	−0.004*** (−3.94)	−0.004*** (−3.92)
Amihud _{<i>d-1</i>}	0.157 (1.22)	0.384*** (2.83)	0.383*** (2.81)
Spread _{<i>d-1</i>}	−0.033* (−1.78)	−0.031* (−1.72)	−0.031* (−1.70)
Mid-Q Return _{<i>d-1</i>}	−0.137*** (−3.62)	−0.163*** (−4.33)	−0.164*** (−4.35)
Mid-Q Return _{<i>d-2</i>}	0.005 (0.14)	−0.028 (−0.77)	−0.030 (−0.82)
Mid-Q Return _{<i>d-3</i>}	−0.034 (−0.93)	−0.062* (−1.68)	−0.063* (−1.71)
Mid-Q Return _{<i>d-4</i>}	−0.051 (−1.30)	−0.072* (−1.86)	−0.072* (−1.87)
Mid-Q Return _{<i>d-5</i>}	0.022 (0.56)	0.001 (0.03)	0.000 (0.00)
Firm Fixed Effect	Yes	Yes	Yes
Week Fixed Effect	Yes	Yes	Yes
R-Square	0.244	0.268	0.268
Num of Observations	779	779	779

necessarily imply that hedge funds do not utilize satellite data in their daily trading; instead, it suggests that non-hedge funds may potentially engage in the use of this data.

To test for robustness, this study performs a difference-in-differences analysis and re-estimates the primary results presented in Tables 2, 4 and 6. Specifically, each covered stock is matched with three non-covered stocks that (1) do not experience coverage by RS Metrics, (2) belong to the same industry, and (3) have the closest market capitalization to the corresponding covered stocks.⁴ Table 7 demonstrates that the results remain consistent even when the difference-in-differences approach is employed.

5. Conclusion

This study examines satellite data's impact on the return predictability of institutional trading. Analyzing data from RS Metrics and ANcerno provides clear evidence that the introduction of satellite data significantly improved the return predictability of

⁴ This study does not employ a difference-in-differences analysis as the main approach due to the poor matching quality between the groups of covered and non-covered stocks in the matched sample. Consistent with Bonelli and Foucault (2024), it is observed that large retailer stocks are significantly more likely to be selected for satellite coverage. For example, four of the five largest retailers in the U.S. are covered by satellite data. In such cases, it is nearly impossible to find non-covered stocks with comparable market capitalizations to the largest covered stocks in the same industry, resulting in poor matching quality.

Table 7
Robustness tests.

This table presents the difference-in-differences regression results using the sample of industry-size matched stocks. Each covered stock is matched with three non-covered stocks that (1) do not experience coverage by RS Metrics, (2) belong to the same industry, and (3) have the closest market capitalization to the corresponding covered stock. All variables are the same as defined in Table 1. The sample includes 19 major retailer stocks with data available from RS Metrics, ANcerno, and CRSP. All coefficient estimates are multiplied by 100. Corresponding *t*-statistics are reported in parentheses. ***, **, and * indicate statistical significance at 1, 5, and 10 percent level, respectively.

	(1)	(2)	(3)	(4)	(5)
$\text{Sum}(\text{AN})_{d-5,d-1} \times \text{After}_d \times \text{Treat}_i$	0.020** (2.52)	0.018** (2.34)			
$\text{Sum}(\text{AN})_{d-5,d-1} \times \text{After}_d \times \text{Treat}_i \times \text{Earnings}_d$		0.070*** (2.67)			
$\text{Sum}(\text{AN})_{d-5,d-1} \times \text{After}_d$	-0.008 (-1.38)	-0.008 (-1.40)			
$\text{Sum}(\text{AN})_{d-5,d-1} \times \text{Treat}_i$	-0.010* (-1.69)	-0.011* (-1.79)			
$\text{Sum}(\text{AN})_{d-5,d-1}$	0.003 (0.65)	0.003 (0.71)			
$\text{Sum}(\text{AN HF})_{d-5,d-1} \times \text{After}_d \times \text{Treat}_i$			-0.208 (-0.39)		-0.413 (-0.76)
$\text{Sum}(\text{AN HF})_{d-5,d-1} \times \text{After}_d$			0.458** (2.12)		0.520** (2.38)
$\text{Sum}(\text{AN HF})_{d-5,d-1} \times \text{Treat}_i$			0.455 (1.45)		0.549* (1.74)
$\text{Sum}(\text{AN HF})_{d-5,d-1}$			-0.423** (-2.25)		-0.438** (-2.32)
$\text{Sum}(\text{AN NHF})_{d-5,d-1} \times \text{After}_d \times \text{Treat}_i$				0.020** (2.56)	0.021*** (2.66)
$\text{Sum}(\text{AN NHF})_{d-5,d-1} \times \text{After}_d$				-0.008 (-1.43)	-0.009 (-1.58)
$\text{Sum}(\text{AN NHF})_{d-5,d-1} \times \text{Treat}_i$				-0.010* (-1.74)	-0.011* (-1.80)
$\text{Sum}(\text{AN NHF})_{d-5,d-1}$				0.003 (0.71)	0.003 (0.69)
Control Variables	Yes	Yes	Yes	Yes	Yes
Firm Fixed Effect	Yes	Yes	Yes	Yes	Yes
Week Fixed Effect	Yes	Yes	Yes	Yes	Yes
R-Square	0.113	0.115	0.112	0.113	0.115
Num of Observations	2,878	2,878	2,878	2,878	2,878

institutional trading, particularly for stocks with high information asymmetry. Furthermore, the results reveal that non-hedge funds also contributed to this improvement. A robustness check using the difference-in-differences approach corroborates these results by matching covered stocks with non-covered counterparts within the same industry that possess similar market capitalizations. These results underscore the increasing influence of alternative data in financial markets and suggest the need for further exploration of its role in areas such as oil shocks or climate policies (Dai and Zhang, 2023; Dai and Wu, 2024).

Data availability

The authors do not have permission to share data.

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